Photovoltaic Market Analysis: India's Renewable Energy: An In-depth Study of India's Solar PV Sector

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Executive Summary

This report provides an in-depth analysis of the photovoltaic (PV) market in India, focusing on investment trends, energy demand, and the role of renewable energy. It also discusses the global energy sector's investment patterns, the energy security risk posed by reliance on a few suppliers, and the impact of these factors on India's PV market.

1.1 Global Energy Investment Trends

In Q1 2025, the global energy sector adopted a cautious approach to major new investments, particularly those dependent on international supply chains. The United States announced around USD 9 billion of new clean manufacturing investments, but nearly USD 7 billion were cancelled, primarily battery projects that rely on imports of critical minerals.

1.2 Energy Security Risks

The energy security risk posed by reliance on a few suppliers for imports of fuels and energy technologies is a significant concern. This was underscored by the European Union's dependence on gas imports from Russia, which exposed vulnerabilities during the global energy crisis of 2022.

1.3 India's Energy Demand and Investment Trends

India's electricity demand has been increasing sharply due to factors such as growth in commercial and residential spaces, increased ownership of air conditioners and appliances, and rising industrial demand. Over the past five years, India has seen the third-largest growth in power generation capacity in the world, following China and the United States. Investment in renewables, particularly solar PV, has surged, constituting over half of total non-fossil investment during this period. In 2024, 83% of power sector investment in India was directed towards clean energy. The country also received around USD 2.4 billion in Development Finance Institution (DFI) funding for clean energy generation projects in 2024, making it the largest recipient of such funding globally.

1.4 Foreign Direct Investment (FDI) in India's Energy Sector

While domestic sources meet a large share of the investment in India's power generation capacity and transmission networks, FDI has been growing steadily, reaching USD 5 billion in 2023, nearly double the pre-pandemic levels. This growth is partly due to rules allowing 100% FDI across electricity generation sources (except nuclear) and transmission infrastructure. However, foreign portfolio investment in energy has declined in the past two years due to various macroeconomic and sectoral factors.

1.5 Challenges and Opportunities in India's PV Market

Despite the investment growth, India's PV market faces several challenges. These include the debt of power distribution companies (discoms), which owed over $9 billion in unpaid dues to renewable energy generators as of March 2025, and the lack of sufficient power storage systems to store excess energy generated during peak times. However, India is also capitalizing on opportunities to become a major player in the solar PV manufacturing industry. The country is implementing a series of tariff and

non-tariff barriers to encourage domestic production and is capitalizing on government contracts to install rooftop solar for 27 million households by the end of the decade, requiring companies to manufacture the panels domestically.

1.6 India's Strategy to Diversify Supply Chains

India is focusing on diversifying its supply chain and leveraging its large domestic demand, which drives around 70% of its GDP. The country's wind and solar capacity has nearly doubled in the past five years, making it the world's third-largest generator of electricity from renewable sources after China and the United States. India's PV market presents a mix of challenges and opportunities. While the country faces issues such as reliance on imports for critical components and the debt of power distribution companies, it is also capitalizing on its large domestic demand and government policies to become a major player in the solar PV manufacturing industry. The country's strategy to diversify its supply chains and leverage its domestic demand could potentially reshape the global PV market in the coming years.

Summary and Trends

Industry Assessment and Scope

2.1 Introduction & Scope

Introduction & Scope This report provides a comprehensive analysis of the photovoltaic (PV) market in India, focusing on investment trends, energy demand, and the role of renewable energy. The scope of the report includes an examination of the global energy sector, the energy security risk posed by reliance on a few suppliers for imports of fuels and energy technologies, and the growth of India's electricity demand. Page of Photovoltaic Market Analysis: India's Renewable Energy The report utilizes a variety of methodologies to gather and analyze data, including a review of relevant literature, analysis of market trends, and examination of investment patterns. The report covers the period from Q1 2025, when the global energy sector saw a "wait and see" approach to major new investments, particularly those dependent on international supply chains. The report highlights the surge in investment in renewables, particularly solar PV, in India over the past five years. It also discusses the role of Foreign Direct Investment (FDI) in India's power generation capacity and transmission networks, and the challenges faced by the sector, including the debt of power distribution companies and

the lack of sufficient power storage systems. The report also examines India's strategy to become a major player in the solar PV manufacturing industry, the country's reliance on Chinese imports for key components of renewable energy technologies, and the efforts to diversify the supply chain. The report also discusses the impact of the Trump administration's trade policies on India's solar module exports to the United States, and the strategies of Indian PV manufacturers to focus on the export market. The report concludes with a projection of India's solar capacity by 2030, the challenges faced by the PV manufacturing sector, and the potential for India to achieve self-sufficiency in solar energy and aggressively target the export market. The report aims to provide a comprehensive understanding of the current state and future prospects of the PV market in India, and to inform strategic decision-making for stakeholders in the sector. The report assumes that the trends and patterns observed during the period under review will continue into the future, barring any major disruptions or changes in policy.

2.2 Industry Overview: Photovoltaic Solar Market - Global and India Context

The global energy sector has been experiencing a cautious approach to major new investments, particularly those dependent on international supply chains. In Q1 2025, around USD 9 billion of new clean manufacturing investments were announced in the United States, but nearly USD 7 billion were cancelled, primarily battery projects that rely on imports of critical minerals. This cautious approach underscores the energy security risk posed by reliance on a few suppliers for imports of fuels and energy technologies. A case in point is the European Union's dependence on gas imports from Russia, which exposed vulnerabilities during the global energy crisis of 2022.

2.3 India's Photovoltaic Solar Market

India's electricity demand has been increasing sharply due to factors such as growth in commercial and residential spaces, increased ownership of air conditioners and appliances, and rising industrial demand. Over the past five years, India has seen the third-largest growth in power generation capacity in the world, following China and the United States. Investment in renewables, particularly solar PV, has surged, constituting over half of total non-fossil investment during this period. In 2024, 83% of power sector investment in India was directed towards clean energy. The country also received around USD 2.4 billion in Development Finance Institution (DFI) funding for clean

energy generation projects in 2024, making it the largest recipient of such funding globally.

2.4 Foreign Direct Investment (FDI) in India's Clean Energy Sector

While domestic sources meet a large share of the investment in India's power generation capacity and transmission networks, FDI has been growing steadily, reaching USD 5 billion in 2023, nearly double the pre-pandemic levels. This growth is partly due to rules allowing 100% FDI across electricity generation sources (except nuclear) and transmission infrastructure. However, foreign portfolio investment in energy has declined in the past two years due to various macroeconomic and sectoral factors.

2.5 Challenges and Opportunities

Despite the surge in investment and growth, the Indian PV solar market faces challenges such as grid limitations and financial risks, which Page of Photovoltaic Market Analysis: India's Renewable Energy have slowed the deployment of clean energy projects. Investor confidence is affected by the debt of power distribution companies (discoms), which owed over $9 billion in unpaid dues to renewable energy generators as of March 2025. To counter these challenges, India introduced new tenders, including hybrid and firm and dispatchable renewables. Round-the-clock (RTC) tenders, battery energy storage integration, and improvements in payment security mechanisms have also contributed to the increase in FDI.

2.6 India's Solar Capacity and Manufacturing

India's solar capacity is projected to reach 292 GW by 2030, nearly double the pre-COVID-19 levels. The country has been implementing a series of tariff and non-tariff barriers since 2018, including a safeguard duty and basic customs duty, to encourage domestic production. Despite increasing its solar panel production, India still relies heavily on Chinese imports for the cells that go into the panels. To counter this, India is focusing on diversifying its supply chain and leveraging its large domestic demand, which drives around 70% of its GDP.

2.7 India's Export Market

In the previous year, over half of India's solar modules were exported to the United States. The Trump administration has imposed lower duties on goods imported from

India (27%) compared to those from China (145%) and Southeast Asia (up to 3,500%). This shift has made India the highest exporter, prompting renewable energy companies to consider whether to focus on the Indian market or sell Indian-made goods abroad.

2.8 Conclusion

India's PV solar market has shown significant growth and potential, driven by increasing domestic demand, supportive government policies, and a surge in foreign direct investment. However, challenges such as grid instability, financial risks, and overreliance on imports for key components persist. As India continues to diversify its supply chain and increase domestic production, the country is poised to become a significant player in the global PV solar market.

3.1 Introduction

India's PV market has been experiencing significant growth, with the country's electricity demand increasing sharply due to factors such as growth in commercial and residential spaces, increased ownership of air conditioners and appliances, and rising industrial demand. Over the past five years, India has seen the third-largest growth in power generation capacity in the world, following China and the United States. In 2024, 83% of power sector investment in India was directed towards clean energy. The country received around USD 2.4 billion in Development Finance Institution (DFI) funding for clean energy generation projects, making it the largest recipient of such funding globally. India's solar capacity is projected to reach 292 GW by 2030, nearly double the pre-COVID-19 levels. Investment in renewables, particularly solar PV, has surged, constituting over half of total non-fossil investment during this period. The rapid deployment of solar PV technologies has raised concerns around solar waste management. However, the decline in solar PV module prices and increased investments have made solar power the cheapest source of clean electricity globally. India's solar capacity has grown 26 times in the last decade, reaching 73.32 GW in December 2023. The Indian government aims to install 500 GW of non-fossil fuel capacity by 2030, having already achieved 234 GW.

3.2 Residential Rooftop vs Utility-Scale Demand Drivers

India has been implementing a series of tariff and non-tariff barriers since 2018 to encourage domestic production. The country is also capitalizing on government contracts to install rooftop solar for 27 million households by the end of the decade, requiring companies to manufacture the panels domestically. On the other hand, utility-scale solar projects are cheaper to build than operating existing coal plants. In 2015, India set a target of installing 175 GW of renewable energy (RE) capacity by 2022, with 100 GW from solar photovoltaics (PV), including 60 GW from utility-scale solar and 40 GW from rooftop solar.

3.3 Foreign Direct Investment (FDI) and Domestic Investment

While domestic sources meet a large share of the investment in India's power generation capacity and transmission networks, FDI has been Page of Photovoltaic Market Analysis: India's Renewable Energy growing steadily, reaching USD 5 billion in 2023, nearly double the pre-pandemic levels. This growth is partly due to rules allowing 100% FDI across electricity generation sources (except nuclear) and transmission infrastructure. However, foreign portfolio investment in energy has declined in the past two years due to various macroeconomic and sectoral factors. Despite this, India was the largest recipient of global development finance funding for energy in 2024, receiving around $2.4 billion for clean energy projects.

3.4 Challenges and Opportunities

Despite the growth and potential, India's PV market faces several challenges. These include grid limitations, financial risks, the debt of power distribution companies, and the lack of sufficient power storage systems. However, measures such as round-the-clock (RTC) tenders, battery energy storage integration, and improvements in payment security mechanisms have been introduced to address these issues. Furthermore, India's strategy to become a major player in the solar photovoltaic (PV) manufacturing industry presents significant opportunities. The country is aiming to satisfy its own energy demands and provide for other countries looking to diversify their energy supply

chains. This includes focusing on diversifying its supply chain and leveraging its large domestic demand, which drives around 70% of its GDP. India's PV market is growing rapidly, driven by increasing electricity demand, investment in renewables, and government initiatives. While there are challenges to overcome, the opportunities presented by the market are significant. With the right strategies and policies in place, India is well-positioned to become a major player in the global PV market.

3.5 Domestic Manufacturing Landscape

India's photovoltaic (PV) manufacturing sector has been experiencing significant growth, driven by increasing domestic electricity demand, aggressive capacity auctions, and a surge in foreign direct investment (FDI). This report provides an overview of the current domestic PV manufacturing players, their capacities, recent expansions, and key clusters and states. Several key players, including Waaree Energies, Adani Solar, and Vikram Solar dominate India's PV manufacturing landscape. These companies were the country's largest PV exporters in FY2024, exporting over 50% of their output. Other significant players in the sector include Grew Energy, ReNew Power, Navitas, Solex Energy, and Saatvik Energy. These companies are not only expanding their domestic operations but also setting up supply chains in international markets.

3.6 Manufacturing Capacities

India's solar PV module capacity is projected to reach 110 GW by 2026, according to the Institute for Energy Economics and Financial Analysis (IEEFA) and JMK Research. This increase in capacity is expected to enable India to achieve self-sufficiency in solar energy and aggressively target the export market. However, the country still relies heavily on imports for upstream components such as polysilicon, ingots/wafers, ancillaries, and PV machinery, particularly from China.

3.7 Recent Expansions

Several Indian PV manufacturers are expanding their operations both domestically and internationally. For instance, Waaree Energies and Vikram Solar are planning to

establish PV manufacturing capacities in the U.S., leveraging incentives under the Inflation Reduction Act (IRA). Other manufacturers like Grew Energy, ReNew Power, Navitas, Solex Energy, and Saatvik Energy are also pursuing export markets and setting up supply chains abroad.

3.8 Key Clusters and States

While the report does not provide specific details on key clusters and states for PV manufacturing in India, it is known that the country's solar capacity is spread across various states. The government's push for rooftop solar installations for 27 million households by the end of the decade is expected to further drive the growth of PV manufacturing across the country.

3.9 Challenges and Opportunities

Despite the growth in India's PV manufacturing sector, several India's Renewable Energy challenges persist. These include the overreliance on imports for key components, the debt of power distribution companies, and the lack of sufficient power storage systems. However, the government's focus on diversifying the supply chain, leveraging domestic demand, and introducing new subsidies for locally produced solar cells and battery cells presents significant opportunities for the sector.

3.10 Conclusion

India's PV manufacturing sector is poised for significant growth, driven by increasing domestic demand, government support, and rising FDI. However, the sector must address several challenges, including overreliance on imports and grid instability, to fully realize its potential. With the right strategies and policies, India has the potential to become a major player in the global PV manufacturing industry

Market Size and Capacity

4.1 Overview

The photovoltaic (PV) market in India has seen significant growth over the past five years, driven by increasing electricity demand and a surge in investment in renewable energy. However, the supply chain for PV manufacturing, which includes sourcing of polysilicon, ingots, wafers, cells, modules, Balance of System (BOS), inverters, glass, silver, and Ethylene Vinyl Acetate (EVA), faces several challenges, including sourcing bottlenecks and logistics issues.

4.2 Sourcing

India's PV manufacturing sector is heavily reliant on imports for upstream components such as polysilicon, ingots/wafers, ancillaries, and PV machinery. Over 90% of the polysilicon used in solar panels is controlled by China, and Indian companies that manufacture solar cells typically import silicon wafers from China. This overreliance on imports poses a significant risk to the supply chain, particularly in light of the global energy crisis of 2022, which underscored the vulnerabilities of relying on a few suppliers for imports of fuels and energy technologies.

4.3 Bottlenecks

The dependence on imports also creates bottlenecks in the supply chain. For instance, if China redirects its excess supply to India at lower prices to compensate for US losses, domestic manufacturers may face margin pressures and price competition. Furthermore, the debt of power distribution companies (discoms), which owed over $9 billion in unpaid dues to renewable energy generators as of March 2025, affects investor confidence and slows down the deployment of clean energy projects.

4.4 Logistics

Logistics issues also pose a challenge to the supply chain. The lack of sufficient power storage systems to store excess energy generated during peak times reduces the effectiveness and reliability of renewable energy. Discoms often fail to honour power

purchase agreements, turning to costlier or less efficient alternatives, which deters investor confidence.

4.5 Mitigation Strategies

To mitigate these challenges, India has been implementing a series of tariff and non-tariff barriers since 2018, including a safeguard duty and basic customs duty, to encourage domestic production. The country is also capitalizing on government contracts to install rooftop solar. India's Renewable Energy for 27 million households by the end of the decade, requiring companies to manufacture the panels domestically. Furthermore, India is focusing on diversifying its supply chain and leveraging its large domestic demand, which drives around 70% of its GDP. The country has been offering subsidies for locally produced solar panels, which are now being discontinued. However, new subsidies are set to be introduced next year for locally produced solar cells and battery cells.

4.6 Future Outlook.

Despite the challenges, the future outlook for India's PV market is positive. The country's solar capacity is projected to reach 292 GW by 2030, nearly double the pre-COVID-19 levels, driven by rules allowing 100% foreign direct investment (FDI) in electricity generation and transmission infrastructure, excluding nuclear. The country's strategy to become a major player in the solar PV manufacturing industry, aiming to satisfy its own energy demands and to provide for other countries looking to diversify their energy supply chains, is also expected to drive growth in the sector. In conclusion, while the supply chain for PV manufacturing in India faces several challenges, the country's focus on encouraging domestic production, diversifying its supply chain, and leveraging its large domestic demand, along with the positive future outlook for the PV market, provide a strong foundation for overcoming these challenges and ensuring the continued growth of the sector.

Supply Chain and Inputs

Policy and Incentives

5.1 Government Policies and Incentives

The Indian government has been proactive in implementing policies and incentives to boost the photovoltaic (PV) market. The Production Linked Incentive (PLI) scheme, customs duties, and import duties have been instrumental in encouraging domestic production and reducing reliance on imports. The government's ambitious target of installing 500 GW of non-fossil fuel capacity by 2030, of which a significant portion is expected to be solar PV, is a testament to its commitment to renewable energy. The government has also allowed 100% Foreign Direct Investment (FDI) in electricity generation and transmission infrastructure, excluding nuclear. This policy has resulted in a steady increase in FDI, reaching USD 5 billion in 2023, nearly double the pre-pandemic levels. However, foreign portfolio investment in energy has declined in the past two years due to various macroeconomic and sectoral factors.

5.2 Tender Rules and Local Content Requirements

To address the challenges of integrating variable renewable energy into the grid, India has introduced new tenders, including hybrid and firm and dispatchable renewables. Round-the-clock (RTC) tenders, battery energy storage integration, and improvements in payment security mechanisms have also been implemented. These measures have contributed to the increase in FDI and have helped ensure a consistent power supply. The government has also leveraged its contracts to install rooftop solar for 27 million households by the end of the decade, requiring companies to manufacture the panels domestically. This move is expected to boost local manufacturing and reduce dependence on imports.

5.3 Challenges and Opportunities

Despite these positive developments, the Indian PV market faces several challenges. The debt of power distribution companies (discoms), which owed over $9 billion in unpaid dues to renewable energy generators as of March 2025, has affected investor confidence. The lack of sufficient power storage systems to store excess energy generated during peak times also reduces the effectiveness and reliability of renewable energy. However, India's strategy to become a major player in the solar PV manufacturing industry presents significant opportunities. The country is focusing on diversifying its supply chain and leveraging its large Page of Photovoltaic Market Analysis: India's Renewable Energy domestic demand, which drives around 70% of its

GDP. The government's decision to introduce new subsidies next year for locally produced solar cells and battery cells is expected to further boost the domestic PV market.

5.4 Localisation and Export Potential

India's PV manufacturers, including Waaree Energies, Adani Solar, and Vikram Solar, are increasingly focusing on the export market. These companies accounted for most of India's PV exports until FY2024, with each exporting more than half of its annual production. Other manufacturers like Grew Energy, ReNew Power, Navitas, Solex Energy, and Saatvik Energy are also pursuing export markets and setting up supply chains abroad. The shift in the U.S.'s import duties, which are lower for goods imported from India compared to those from China and Southeast Asia, has made India the highest exporter of solar modules to the U.S. This shift, coupled with the potential imposition of high tariffs on PV products from Southeast Asia by the U.S., presents a significant opportunity for Indian PV manufacturers to increase their exports to the U.S.

5.5 Conclusion

India's PV market is at a critical juncture, with significant opportunities and challenges. The government's policies and incentives, coupled with the country's strategy to become a major player in the solar PV manufacturing industry, present a promising outlook. However, addressing the challenges related to discoms' debt and power storage systems will be crucial for the sustainable growth of the mark

6.1 Manufacturing CAPEX/OPEX Drivers

India's photovoltaic (PV) market has seen significant growth, driven by increasing electricity demand, a surge in investment in renewables, and a steady rise in Foreign Direct Investment (FDI). The country's commitment to clean energy, as evidenced by the 83% of power sector investment directed towards clean energy in 2024, has been a significant driver for manufacturing capital expenditure (CAPEX). The operational expenditure (OPEX) in the PV sector is influenced by factors such as grid limitations, financial risks, and the debt of power distribution companies (discoms). The lack of

sufficient power storage systems and the failure of discoms to honor power purchase agreements also contribute to the OPEX.

6.2 Levelized Cost Implications

The levelized cost of electricity (LCOE) from solar PV has been decreasing due to the decline in solar PV module prices and increased investments. This has made solar power the cheapest source of clean electricity globally. However, the cost-effectiveness of solar PV is affected by the challenges of integrating variable renewable energy into the grid and the debt of discoms.

6.3 Break-Even Analysis

Several factors, including the cost of manufacturing, the price of solar PV modules, and the level of investment in the sector influence the break-even point for India's PV sector. The increase in FDI, coupled with the funding from Development Finance Institutions (DFIs), has helped lower the break-even point. However, the financial risks associated with the sector, including the debt of discoms, could potentially increase the break-even point.

6.4 Comparison with Imports/Chinese Products

India has been reducing its reliance on Chinese solar modules and PV cells. The share of Chinese PV cells and modules in India's imports decreased from over 90% in FY2022 to 56% and 65% respectively in FY2024. Despite this, India still depends on China for over 50% of its PV cell and module components, particularly polysilicon and wafers. The Indian government has implemented tariff and non-tariff barriers to encourage domestic production and reduce dependence on Chinese imports. However, the dominance of China in the production of key components for renewable energy technologies poses a challenge.

6.5 Conclusion

India's PV market is growing rapidly, driven by increasing electricity demand, a surge in investment in renewables, and a steady rise in FDI. However, the sector faces challenges, including grid limitations, financial risks, and the debt of discoms. Despite these challenges, India is making strides in reducing its reliance on Chinese imports and encouraging domestic production. The country's commitment to clean energy and the increasing level of investment in the sector suggest a positive outlook for the PV market in India.